Reporting Regulation, Market Liquidity, and Policy Diffusion

Abstract

We study reporting regulation in a global market sample of 90 jurisdictions over 2000-2025. First, we compare country-average stock liquidity around national implementation. Seventeen policy families are considered; 13 support following-year comparisons. The spread estimate is +0.007 with equal family weights and +0.018 with market-cap weights. Both intervals include zero. Second, we examine competition, learning, emulation and external finance as explanations for policy adoption. Historical data support broader separate tests, but the joint model covers only one family. The evidence does not establish a common liquidity improvement or a general diffusion mechanism across the expanded policy set.

Introduction

Corporate reporting can help investors compare companies and trade on more equal terms. Its benefits depend on what companies disclose, who receives the information and how rules are enforced. Public sustainability reports and confidential tax returns therefore need not affect trading in the same way. We ask two questions: how does liquidity change around national regulation, and which international connections are associated with adoption?

Krueger et al. (2023) study mandatory ESG disclosure and liquidity. We extend the setting to 17 policy families, using country averages to retain international coverage. Koning, Mertens and Roosenboom (2018) study the diffusion of IFRS; we adapt their mechanisms to the expanded policy panel. These extensions change both the economic setting and the available comparisons, so agreement across policies is an empirical question.

1. Liquidity

Datastream contains 1,055,142 stock-year observations for 99,100 vendor company records in 90 jurisdictions over 2000-2025. All three liquidity measures are available for 636,960 stock-years in 86 jurisdictions. We measure bid-ask spreads, the median daily price response to trading volume, and the fraction of observed valid daily returns equal to zero. Higher values indicate less liquid trading. These records are not a census of all listed companies; each comparison uses a smaller fixed company sample.

Table 1 distinguishes the policy families and their supported cohorts. We preserve the difference between the start of a covered financial period and the arrival of public information. For UK extended audit reports, the standard was issued in 2013 but covered periods beginning from October 2012; the information-policy clock uses 2013. The broader ESG register often describes a later reporting phase. A separate benchmark uses the complete-topic mandatory dates in Krueger et al. through 2020, including comply-or-explain mandates. Neither calendar supplies actual filing dates for every company.

For each family and adoption cohort, we compare changes in country-average liquidity with changes in countries whose documented adoption comes after the outcome window. Unknown histories and already-treated countries do not supply controls. Following the outcome timing in Krueger et al., the main comparison uses liquidity in year 1 after the policy year, relative to year -1. We also average years 1-3. Companies and share lines are held fixed within a cohort; countries need 30 companies with complete outcomes and cohorts need three comparison countries. The two horizons support 13 and 10 families, respectively.

Prior-year GDP weights countries and adopting cohorts within each family. Across families, equal weights describe the average supported family; market-cap weights give more influence to families with larger sampled pre-implementation company values. These are different economic averages. They use the same family estimates, but companies can appear under several policies. The market-cap pool is not a portfolio of distinct world companies, and complete outcome requirements also determine which exposures receive weight.

Table 2 reports both weights. The following-year spread estimate is +0.007 with equal weights and +0.018 with market-cap weights; both are imprecise. Formal comparisons do not establish that the two weighted effects differ. Figure 1 shows why the point estimates move: confidential tax country reporting receives 36.1% of market-cap weight, compared with 7.7% under equal weights. Auditor reporting receives 10.8%. These exposure differences do not establish a disclosure benefit for investors.

We retain all 1,296 declared specification checks across calendars, outcomes, weights, horizons, policy groups, coordinated adoptions and controls. 837 estimates are negative and 459 positive; 40 pass the grid-wide Holm correction. The choices were declared after prior results were known. Correlated model fits are not independent confirmations, and selected significant cells do not replace the main tests. Figure 2 also shows differences before implementation on the balanced event sample. Together, these results limit a claim that the expanded policy set causes a common liquidity improvement.

2. Policy diffusion

We next ask whether countries follow foreign peers. As in Koning et al., the main outcome records whether a policy is in place; a separate sensitivity studies first-adoption timing. We include family and year effects and exclude coordinated focal adoptions from the main models. Competition measures earlier adoption among five export-market peers. Learning compares lagged GDP growth among geographic adopters and nonadopters. Emulation uses language peers as a proxy for cultural proximity. External finance measures IMF credit and aid relative to GDP.

The learning measure equals zero when no neighbour has adopted. When all have adopted, their median growth is compared with the focal country. All signals precede the focal outcome. Because many world neighbours have unknown policy histories, the main adaptation selects the ten nearest countries with documented histories. The appendix also retains the stricter world-neighbour definition, annual trade peers through their available window, legal-origin peers, domestic controls and the original 2002-2012 period. These are adaptations, not an exact replication of the JIBS country universe or cultural-distance measure.

Table 3 reports separate mechanisms on their available samples, then all four together. Separate competition and learning models cover seven and eight families; the emulation and external-finance fits are unstable and are withheld. The joint model leaves 90 observations in 15 countries, all for confidential tax country reporting. Equal and market-cap weights must then coincide. These documented phase histories are not a verified global census of adoption and nonadoption. The narrow joint sample cannot establish diffusion across the expanded panel. None of the principal associations survives adjustment for eight comparisons.

References

Krueger, P., Z. Sautner, D. Y. Tang and R. Zhong. 2023. The Effects of Mandatory ESG Disclosure Around the World. Supplied working paper.

Koning, M., G. Mertens and P. Roosenboom. 2018. Drivers of institutional change around the world: The case of IFRS. Journal of International Business Studies.

Policy sources, complete specification ledger, weights and code: https://hunternbh.github.io/policy-diffusion/ . Licensed company observations remain local.

Table 1. International policy coverage

Regulation family	Adopting countries	Cohort years	Cohorts	Market-cap weight	Years 1-3
Audit institutions / rotation	4	2004-2007	3	2.9%	Yes
Auditor reporting	20	2013-2016	2	10.8%	Yes
Beneficial ownership	11	2016-2020	5	11.1%	Yes
Public bank country reports	14	2014	1	7.1%	Yes
Confidential tax country reports	42	2016-2019	4	36.1%	Yes
Public multinational country reports	-	-	-	-	No
Climate / emissions disclosure	2	2013-2023	2	2.2%	Yes
Sectoral climate rules	-	-	-	-	No
Digital-asset regulation	2	2018-2020	2	0.6%	Yes
Nonfinancial reporting	22	2002-2017	9	11.5%	Yes
Payments to governments	1	2014	1	0.2%	No
Corporate governance	5	2001-2003	3	1.2%	No
IFRS financial reporting	20	2005-2010	5	6.1%	Yes
Supply-chain reporting	4	2012-2019	4	10.0%	Yes
Financial-focus sustainability	1	2024	1	0.2%	No
Sustainability frameworks	-	-	-	-	No
Electronic financial reporting	-	-	-	-	No

Following-year GDP-weighted comparisons: 13 supported families; 10 support years 1-3. Future schedules are excluded. Counts describe national exposure, not verified covered issuers. A dash denotes no supported comparison, not no policy. Dates refer to documented phases; nonfinancial phases need not be the first complete E, S and G mandate. Public bank phases enter once.

Dates and interpretation

National phase dates do not identify the first public report for every company. The reviewed register differs from the supplied ESG adoption calendar in 25 of 31 overlapping dated countries. That difference reflects scope and timing conventions as well as possible history gaps; it is not 25 established legal errors. The published ESG calendar is tested separately through its sample endpoint, rather than added as a second policy family. Unknown histories remain outside the comparison group. Confidential CbCR is filed with tax authorities and exchanged between them; public bank and public multinational country reports make specified country information publicly available.

Table 2. Liquidity after national implementation

Window and outcome	Equal family Estimate [95% interval]	Market-cap family Estimate [95% interval]
Year 1: 13 families Bid-ask spread (log)	+0.007 [-0.039, +0.052]	+0.018 [-0.050, +0.087]
Year 1: 13 families Price impact (log)	+0.027 [-0.048, +0.101]	-0.025 [-0.189, +0.139]
Year 1: 13 families Unchanged returns (fraction)	-0.0021 [-0.0062, +0.0019]	+0.0037 [-0.0063, +0.0137]
Years 1-3: 10 families Bid-ask spread (log)	-0.022 [-0.091, +0.047]	+0.072 [-0.110, +0.254]
Years 1-3: 10 families Price impact (log)	+0.082 [-0.005, +0.169]	+0.127 [-0.095, +0.350]
Years 1-3: 10 families Unchanged returns (fraction)	-0.0035 [-0.0097, +0.0026]	-0.0003 [-0.0145, +0.0139]

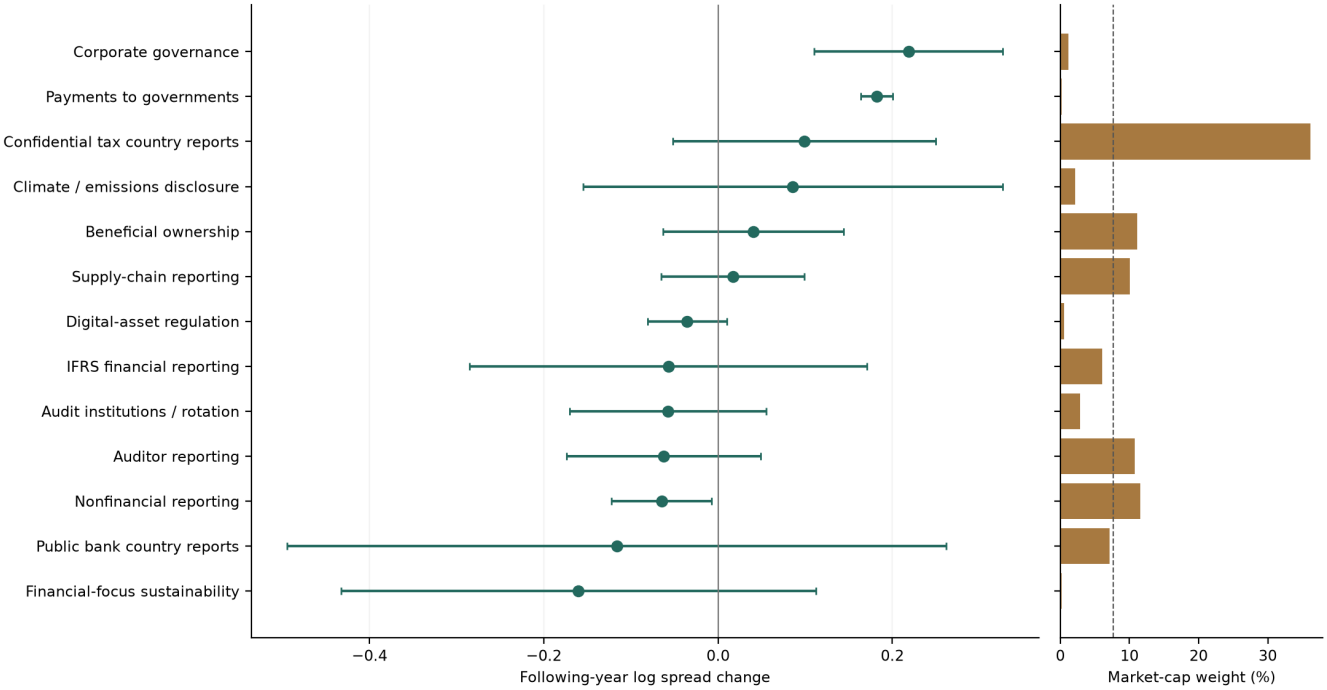
Fixed company cohorts; GDP weights within families. Year -1 is the baseline; the outcome is year 1 or the mean of years 1-3. Higher outcomes indicate lower liquidity. Log points and fractions are not percentages. Intervals describe conditional country sampling uncertainty; they omit some uncertainty from coordinated laws, scope and adoption timing. None of the 12 principal tests survives Holm adjustment. Horizons have different support.

Table 3. International adoption and diffusion

Model and mechanism	Rows / countries / families	Equal family Odds ratio [95% interval]	Market-cap family Odds ratio [95% interval]
Separate: Competition	1,122 / 40 / 7	0.60 [0.37, 0.99]	0.69 [0.46, 1.03]
Separate: Learning	2,674 / 55 / 8	1.02 [0.65, 1.62]	1.23 [0.70, 2.14]
Separate: Emulation	2,424 / 37 / 11	Unstable fit	Unstable fit
Separate: External finance	1,410 / 31 / 9	Unstable fit	Unstable fit
Joint: Competition	90 / 15 / 1	0.62 [0.28, 1.36]	0.62 [0.28, 1.36]
Joint: Learning	90 / 15 / 1	3.04 [0.60, 15.41]	3.04 [0.60, 15.41]
Joint: Emulation	90 / 15 / 1	1.13 [0.57, 2.26]	1.13 [0.57, 2.26]
Joint: External finance	90 / 15 / 1	0.70 [0.48, 1.03]	0.70 [0.48, 1.03]

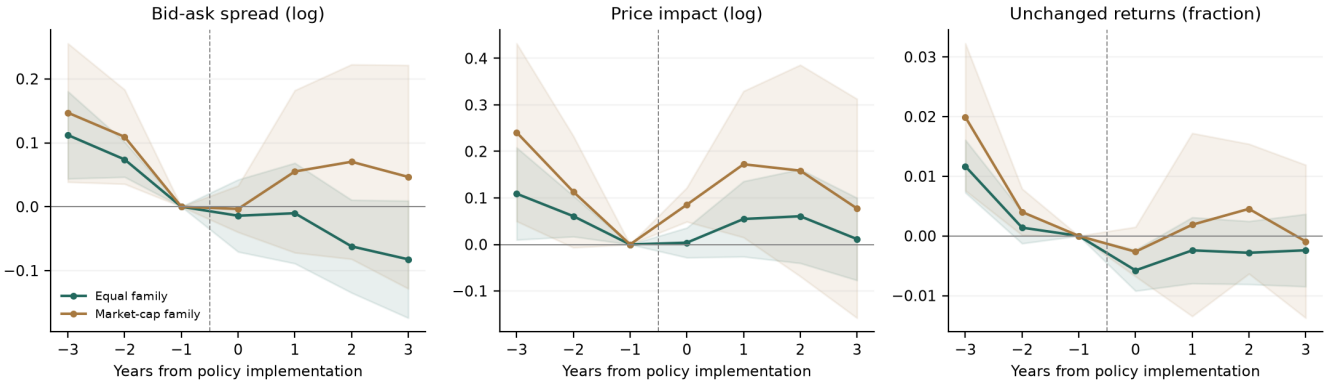
Adoption-status logit with family and year effects; coordinated focal adoptions excluded. Weights come from the 13 liquidity-supported families, restricted to each model's available sample. The joint model leaves confidential tax reporting in 15 countries; its two weights are identical. A ratio above one indicates a positive association. Competition and emulation are per 10 percentage points of peer adoption, learning is binary, and external finance is per one percentage point of GDP. Country-clustered t intervals. Unstable fits are withheld, not zeros. No principal association survives the eight-test Holm adjustment; none of the 44 stable coefficients survives adjustment over the full stable matrix.

Figure 1. Family estimates and market-cap weights



All 13 supported following-year families; GDP weights within families. Bars show 95% conditional country intervals. Dashed weight line is the equal-family weight. Each family has its own supported countries and cohorts.

Figure 2. Liquidity before and after implementation



Nine families with fixed company support in years -3 through +3. Year -1 is the reference. Shading shows pointwise 95% conditional country intervals. This stricter event sample differs from the main 13-family following-year and 10-family longer-window samples. Pre-policy differences and coordinated legal waves limit a causal interpretation.